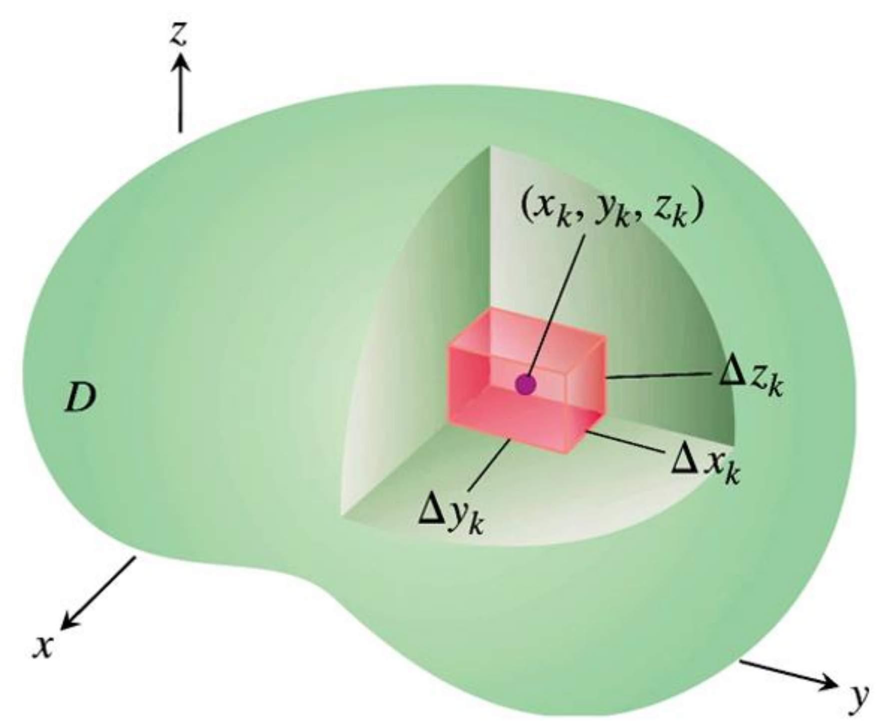
Triple Integrals in Rectangular Coordinates



DEFINITION **Volume**

The **volume** of a closed, bounded region D in space is

$$V = \iiint_D dV.$$

Triple Integrals in Rectangular Coordinates

EXAMPLE 1 Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Solution The volume is

$$V = \iiint_D dz\, dy\, dx,$$

To find the limits of integration for evaluating the integral, we first sketch the region. The surfaces (Figure 15.30) intersect on the elliptical cylinder $x^2 + 3y^2 = 8 - x^2 - y^2$ or $x^2 + 2y^2 = 4$, $z > 0$. The boundary of the region R, the projection of D onto the *xy-plane*, is an ellipse with the same equation: $x^2 + 2y^2 = 4$. The "upper" boundary of R is the curve $y = \sqrt{(4 - x^2)/2}$. The lower boundary is the curve $y = -\sqrt{(4 - x^2)/2}$.

Triple Integrals in Rectangular Coordinates

The line *M* passing through a typical point (x, y) in *R* parallel to the z-axis enters *D* at $z = x^2 + 3y^2$ and leaves at $z = 8 - x^2 - y^2$.

Finally, we find the *x-limits* of integration. As *L* sweeps across *R*, the value of x varies from $x = -2$ at $(-2, 0, 0)$ to $x = 2$ at $(2, 0, 0)$.

Triple Integrals in Rectangular Coordinates

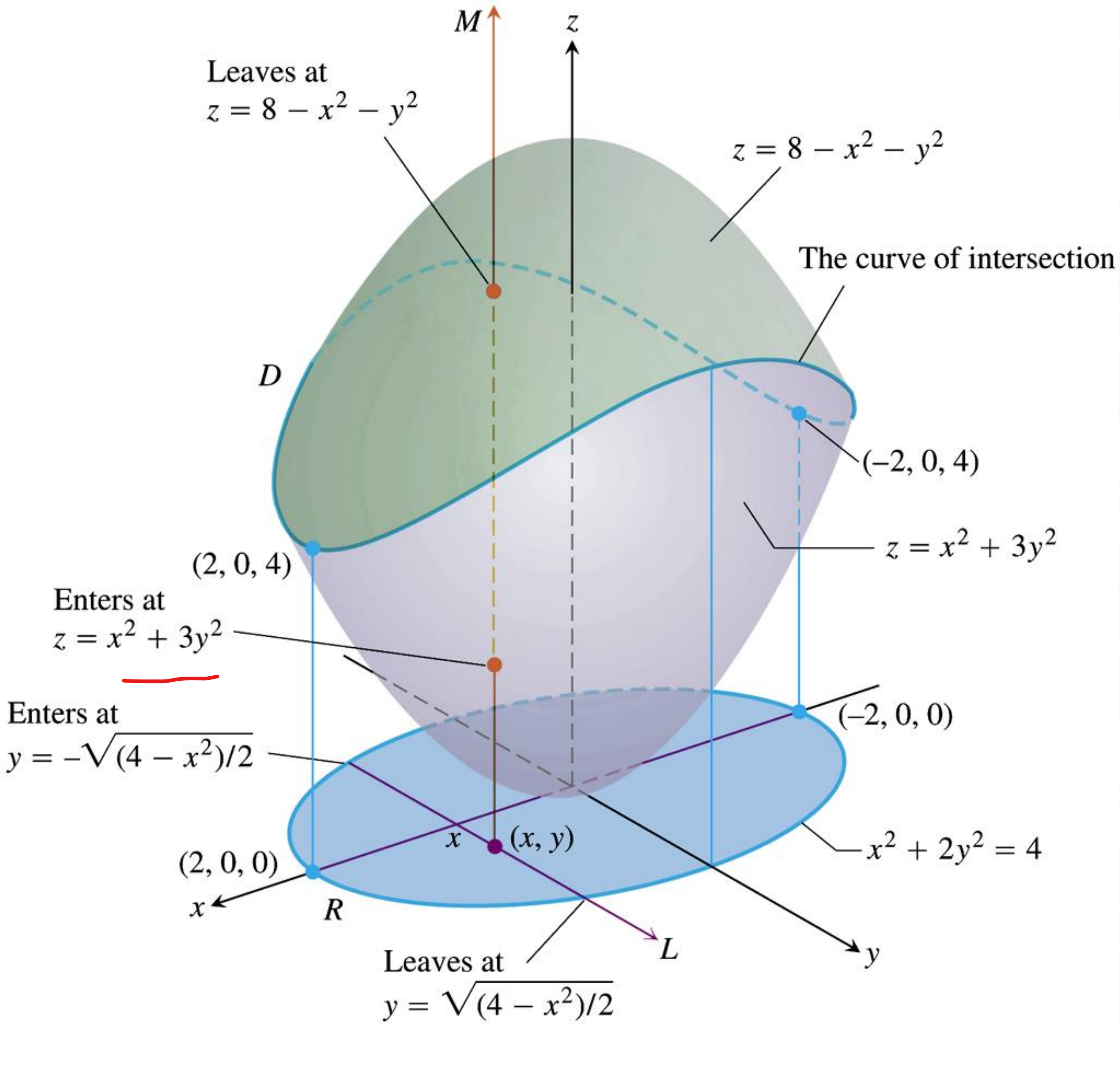


FIGURE 15.28 The volume of the region enclosed by two paraboloids, calculated in Example 1.

Triple Integrals in Rectangular Coordinates

The volume of D *is*

$$V = \iiint_D dz\,dy\,dx$$

$$= \int_{-2}^{2}\int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}}\int_{x^2+3y^2}^{8-x^2-y^2} dz\,dy\,dx$$

$$= \int_{-2}^{2}\int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}}(8-2x^2-4y^2)\,dy\,dx$$

$$= \int_{-2}^{2}\left[(8-2x^2)y-\frac{4}{3}y^3\right]_{y=-\sqrt{(4-x^2)/2}}^{y=\sqrt{(4-x^2)/2}}dx$$

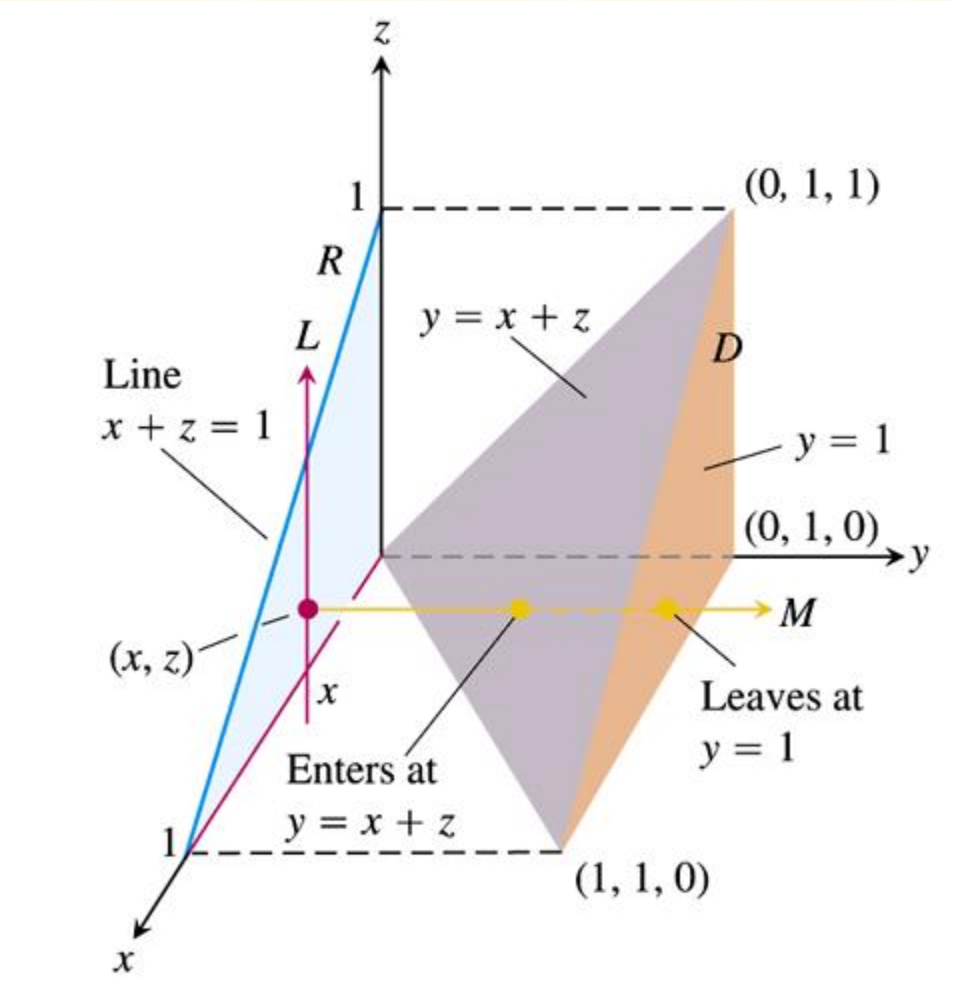
$$= \int_{-2}^{2}\left(2(8-2x^2)\sqrt{\frac{4-x^2}{2}}-\frac{8}{3}\left(\frac{4-x^2}{2}\right)^{3/2}\right)dx$$

$$= \int_{-2}^{2}\left[8\left(\frac{4-x^2}{2}\right)^{3/2}-\frac{8}{3}\left(\frac{4-x^2}{2}\right)^{3/2}\right]dx = \frac{4\sqrt{2}}{3}\int_{-2}^{2}(4-x^2)^{3/2}\,dx$$

$$= 8\pi\sqrt{2}.$$ After integration with the substitution $x = 2\sin u$

Triple Integrals in Rectangular Coordinates

EXAMPLE 2 Integrate $F(x, y, z) = 1$ over the tetrahedron D with vertices (0, 0, 0), (1, 1,0), (0, 1,0), and (0, 1, 1). Use the order of integration $dy\,dz\,dx$.



Triple Integrals in Rectangular Coordinates

First, we find the *y-limits* of integration. The line through a typical point (x, z) in R parallel to the y-axis enters D at $y = x + z$ and leaves at $y = 1$.

Next, we find the z-limits of integration. The line L through (x, z) parallel to the z-axis enters R at $z = 0$ and leaves at $z = 1 - x$.

Finally, we find the *x-limits* of integration. As L sweeps across R, the value of x varies from $x = 0$ to $x = 1$.
The integral is

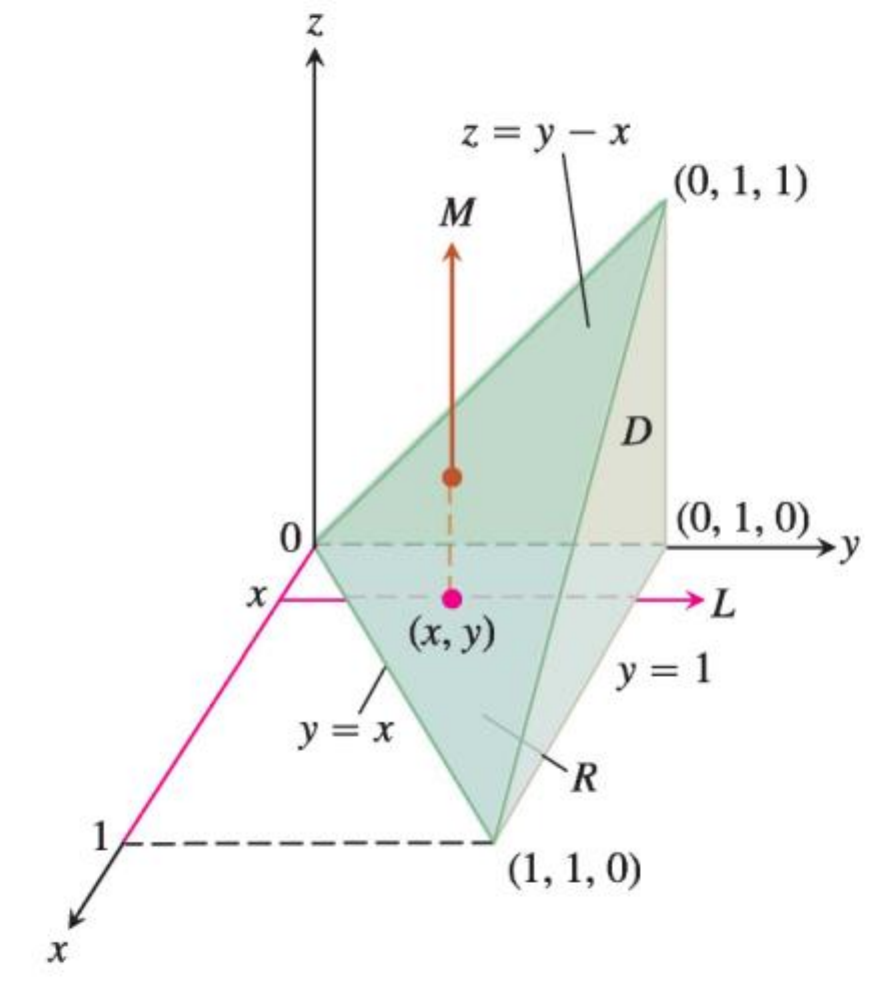
$$\int_0^1 \int_0^{1-x} \int_{x+z}^{1} F(x, y, z)\,dy\,dz\,dx$$

Triple Integrals in Rectangular Coordinates

$$
\begin{aligned}
V &= \int_0^1 \int_0^{1-x} \int_{x+z}^{1} dy\, dz\, dx \\
&= \int_0^1 \int_0^{1-x} (1 - x - z)\, dz\, dx \\
&= \int_0^1 \left[(1 - x)z - \frac{1}{2} z^2 \right]_{z=0}^{z=1-x} dx \\
&= \int_0^1 \left[(1 - x)^2 - \frac{1}{2} (1 - x)^2 \right] dx \\
&= \frac{1}{2} \int_0^1 (1 - x)^2\, dx \\
&= -\frac{1}{6} (1 - x)^3 \Big]_0^1 = \frac{1}{6}.
\end{aligned}
$$

Triple Integrals in Rectangular Coordinates

EXAMPLE 2 Integrate $F(x, y, z) = 1$ over the tetrahedron D with vertices (0, 0, 0), (1, 1,0), (0, 1,0), and (0, 1, 1). Use the order of integration $dzdydx$.



Triple Integrals in Rectangular Coordinates

Solution:

First, we find the z-limits of integration. A line M parallel to the z-axis through a typical point (x, y) in the *xy-plane* "shadow" enters the tetrahedron at $z = 0$ and exits through the upper plane where $z = y - x$ (Figure 15.32).

Next, we find the *y-limits* of integration. On the *xy-plane,* where $z = 0$, the sloped side of the tetrahedron crosses the plane along the line $y = x$. A line L through (x, y) parallel to the y-axis enters the shadow in the *xy-plane* at $y = x$ and exits at $y = 1$ (Figure 15.32).

Finally, we find the x-limits of integration. As the line L parallel to the y-axis in the previous step sweeps out the shadow, the value of x varies from $x = 0$ to $x = 1$ at the point (1, 1, 0) (see Figure 15.32).

Triple Integrals in Rectangular Coordinates

$$
\begin{aligned}
V &= \int_0^1 \int_x^1 \int_0^{y-x} dz\,dy\,dx \\
&= \int_0^1 \int_x^1 (y - x)\,dy\,dx \\
&= \int_0^1 \left[\frac{1}{2}y^2 - xy\right]_{y=x}^{y=1} dx \\
&= \int_0^1 \left(\frac{1}{2} - x + \frac{1}{2}x^2\right) dx \\
&= \left[\frac{1}{2}x - \frac{1}{2}x^2 + \frac{1}{6}x^3\right]_0^1 \\
&= \frac{1}{6}.
\end{aligned}
$$

Triple Integrals in Rectangular Coordinates

Properties of Triple Integrals

If $F = F(x, y, z)$ and $G = G(x, y, z)$ are continuous, then

1. *Constant Multiple:* $\displaystyle\iiint_D kF\,dV = k\iiint_D F\,dV \qquad (\text{any number } k)$

2. *Sum and Difference:* $\displaystyle\iiint_D (F \pm G)\,dV = \iiint_D F\,dV \pm \iiint_D G\,dV$

3. *Domination:*

(a) $\displaystyle\iiint_D F\,dV \geq 0 \qquad \text{if } F \geq 0 \text{ on } D$

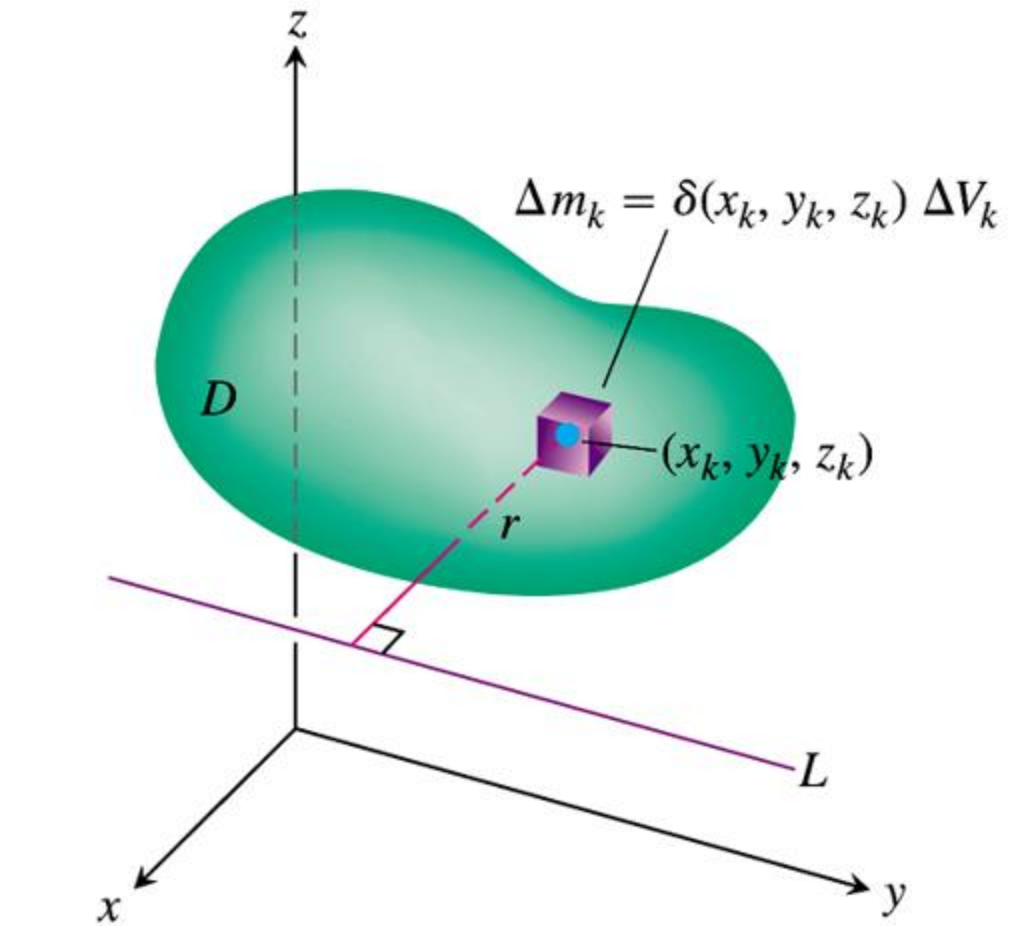
(b) $\displaystyle\iiint_D F\,dV \geq \iiint_D G\,dV \qquad \text{if } F \geq G \text{ on } D$

4. *Additivity:* $\displaystyle\iiint_D F\,dV = \iiint_{D_1} F\,dV + \iiint_{D_2} F\,dV$

if D is the union of two nonoverlapping regions D_1 and D_2.

Masses and Moments in Three Dimensions

$$\text{Mass} = \lim_{n\to\infty} \sum_{k=1}^{n} \Delta m_k = \lim_{n\to\infty} \sum_{k=1}^{n} \rho(xk, yk, zk)\Delta v_k = \iiint_R \rho\left(x_k, y_k, z_k\right) dv$$



Masses and Moments in Three Dimensions

The *first moment* of a solid region D about a coordinate plane is defined as the triple integral over D of the dis1anc from a point *(x, y,* z) in D to the plane multiplied by the density of the solid at that point. For instance, the first moment about the yz-plane is the integral

$$M_{yz} = \iiint_R x\rho(x, y, z)\,dv$$

The *center of mass* is found from the first moments. For instance, the x-coordinate of the center of mass is *x* - *Myz/M*.

$$\overline{x} = \frac{M_{yz}}{M} \iiint_R x\rho(x, y, z)\,dv$$

Masses and Moments in Three Dimensions

Moments of Inertia:

The moments of $M\,x$ and My used in determining the center of mass of a lamina are sometimes called the **first moments** about the x- and y-axes. In each case, the moment is the product of a mass times a distance.

$$M_x = \iint_R \underbrace{(y)}_{\text{Distance to } x\text{-axis}} \underbrace{(x, y)\, dA}_{\text{Mass}} \qquad M_y = \iint_R \underbrace{(x)}_{\text{Distance to } y\text{-axis}} \underbrace{(x, y)\, dA}_{\text{Mass}}$$

You will now look at another type of moment—the **second moment,** or the **moment of inertia** of a lamina about a line. In the same way that mass is a measure of the tendency of matter to resist a change in straight-line motion, the moment of inertia about a line is a *measure of the tendency of matter to resist a change in rotational motion*. For example, when a particle of mass m is a distance d from a fixed line, its moment of inertia about the line is defined as

Masses and Moments in Three Dimensions

$$I = md^2 = (\text{mass})(\text{distance})^2.$$

These second moments are denoted by I_x and I_y, and in each case the moment is the product of a mass times the square of a distance.

$$I_x = \iint_R \underbrace{(y^2)}_{\substack{\text{Square of distance}\\ \text{to } x\text{-axis}}} \underbrace{(x, y)\, dA}_{\text{Mass}} \qquad I_y = \iint_R \underbrace{(x^2)}_{\substack{\text{Square of distance}\\ \text{to } y\text{-axis}}} \underbrace{(x, y)\, dA}_{\text{Mass}}$$

The sum of the moments I_x and I_y is called the **polar moment of inertia** and is denoted by I_o.

$$I_0 = \iint_R (x^2 + y^2)\ (x, y)\, dA = \iint_R (r^2)\ (x, y)\, dA$$

Masses and Moments in Three Dimensions

TABLE 15.3 Mass and moment formulas for solid objects in space

Mass: $M = \iiint_D \delta \, dV \qquad (\delta = \delta(x, y, z) = \text{density})$

First moments about the coordinate planes:

$$M_{yz} = \iiint_D x \, \delta \, dV, \qquad M_{xz} = \iiint_D y \, \delta \, dV, \qquad M_{xy} = \iiint_D z \, \delta \, dV$$

Center of mass:

$$\bar{x} = \frac{M_{yz}}{M}, \qquad \bar{y} = \frac{M_{xz}}{M}, \qquad \bar{z} = \frac{M_{xy}}{M}$$

Moments of inertia (second moments) about the coordinate axes:

$$I_x = \iiint (y^2 + z^2) \, \delta \, dV$$

$$I_y = \iiint (x^2 + z^2) \, \delta \, dV$$

$$I_z = \iiint (x^2 + y^2) \, \delta \, dV$$

Moments of inertia about a line *L*:

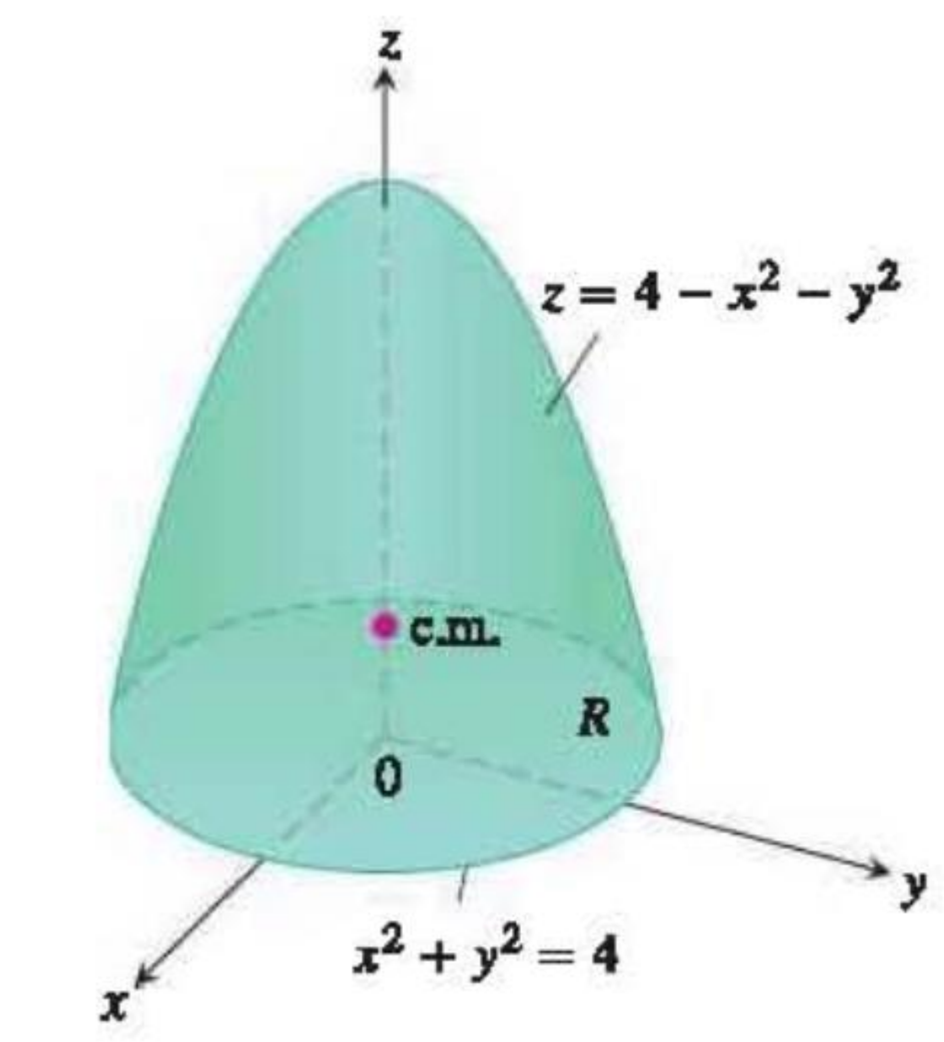
$$I_L = \iiint r^2 \, \delta \, dV \qquad (r(x, y, z) = \text{distance from the point } (x, y, z) \text{ to line } L)$$

Radius of gyration about a line *L*:

$$R_L = \sqrt{I_L / M}$$

Masses and Moments in Three Dimensions

EXAMPLE 1: Find the center of mass of a solid of constant density ρ bounded below by the disk R: $x^2 + y^2 \leq 4$ in the plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$



Masses and Moments in Three Dimensions

EXAMPLE 1 Finding the Center of Mass of a Solid in Space

Find the center of mass of a solid of constant density δ bounded below by the disk $R: x^2 + y^2 \le 4$ in the plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$ (Figure 15.34).

Solution By symmetry $\bar{x} = \bar{y} = 0$. To find $\bar{z}$, we first calculate

$$M_{xy} = \iiint_R^{z=4-x^2-y^2}_{z=0} z\,\delta\,dz\,dy\,dx = \iint_R \left[\frac{z^2}{2}\right]_{z=0}^{z=4-x^2-y^2} \delta\,dy\,dx$$

$$= \frac{\delta}{2}\iint_R (4 - x^2 - y^2)^2\,dy\,dx$$

$$= \frac{\delta}{2}\int_0^{2\pi}\int_0^2 (4 - r^2)^2\,r\,dr\,d\theta \qquad \text{Polar coordinates}$$

$$= \frac{\delta}{2}\int_0^{2\pi}\left[-\frac{1}{6}(4 - r^2)^3\right]_{r=0}^{r=2} d\theta = \frac{16\delta}{3}\int_0^{2\pi} d\theta = \frac{32\pi\delta}{3}.$$

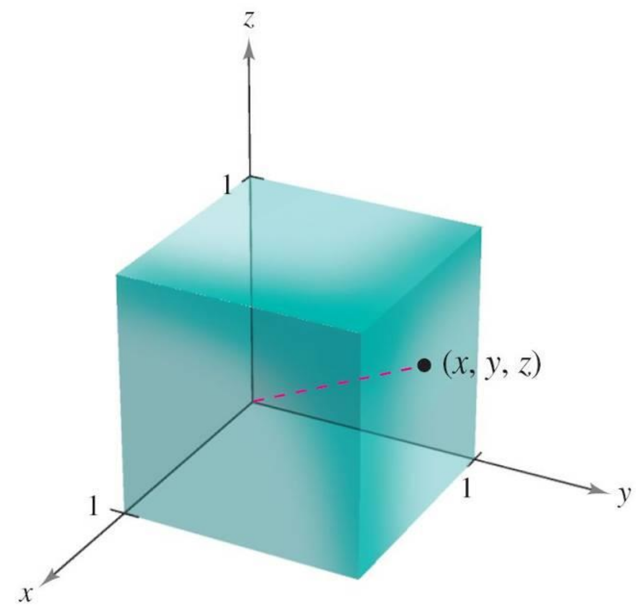
A similar calculation gives

$$M = \iiint_R^{4-x^2-y^2}_{0} \delta\,dz\,dy\,dx = 8\pi\delta.$$

Therefore $\bar{z} = (M_{xy}/M) = 4/3$ and the center of mass is $(\bar{x}, \bar{y}, \bar{z}) = (0, 0, 4/3)$.

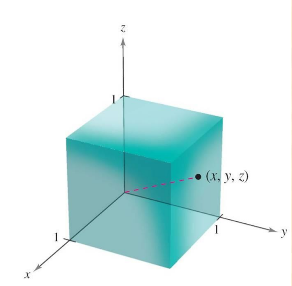
Masses and Moments in Three Dimensions

Find the center of mass of the unit cube shown in the Figure, given that the density at the point (x, y, z) is proportional to the square of its distance from the origin.



Variable density:
$\rho(x, y, z) = k(x^2 + y^2 + z^2)$

Masses and Moments in Three Dimensions



Variable density:
$\rho(x, y, z) = k(x^2 + y^2 + z^2)$

Because the density at (x, y, z) is proportional to the square of the distance between (0, 0, 0) and (x, y, z), you have

$$\boxed{\rho(x, y, z) = k(x^2 + y^2 + z^2).}$$

where k is the constant of proportionality.

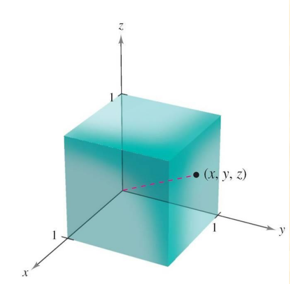
You can use this density function to find the mass of the cube.

Because of the symmetry of the region, any order of integration will produce an integral of comparable difficulty.

$$m = \int_0^1 \int_0^1 \int_0^1 k(x^2 + y^2 + z^2)\,dz\,dy\,dx$$

Apply formula for mass of a solid.

Masses and Moments in Three Dimensions



Variable density:
$\rho(x, y, z) = k(x^2 + y^2 + z^2)$

$$= k\int_0^1\int_0^1 \left[(x^2 + y^2)z + \frac{z^3}{3}\right]_0^1 dy\, dx$$ Integrate with respect to z.

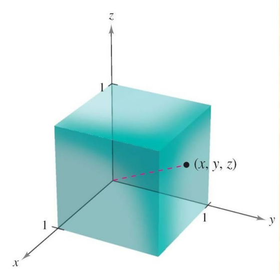
$$= k\int_0^1\int_0^1 \left(x^2 + y^2 + \frac{1}{3}\right) dy\, dx$$

$$= k\int_0^1 \left[\left(x^2 + \frac{1}{3}\right)y + \frac{y^3}{3}\right]_0^1 dx$$ Integrate with respect to y.

$$= k\int_0^1 \left(x^2 + \frac{2}{3}\right) dx$$

$$= k\left[\frac{x^3}{3} + \frac{2x}{3}\right]_0^1$$ Integrate with respect to x. $= k$

Masses and Moments in Three Dimensions



Variable density:
$\rho(x, y, z) = k(x^2 + y^2 + z^2)$

The first moment about the *yz*-plane is

$$M_{yz} = k\int_0^1\int_0^1\int_0^1 x(x^2 + y^2 + z^2)\,dz\,dy\,dx$$

Apply formula for first moment about *yz*-plane.

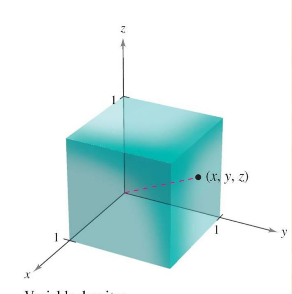
$$= k\int_0^1 x\left[\int_0^1\int_0^1 (x^2 + y^2 + z^2)\,dz\,dy\right]dx.$$

Factor.

Note that x can be factored out of the two inner integrals, because it is constant with respect to y and z.

After factoring, the two inner integrals are the same as for the mass m.

Masses and Moments in Three Dimensions



Therefore, you have

$$M_{yz} = k\int_0^1 x\left(x^2 + \frac{2}{3}\right)dx$$

$$= k\left[\frac{x^4}{4} + \frac{x^2}{3}\right]_0^1$$ Integrate with respect to x.

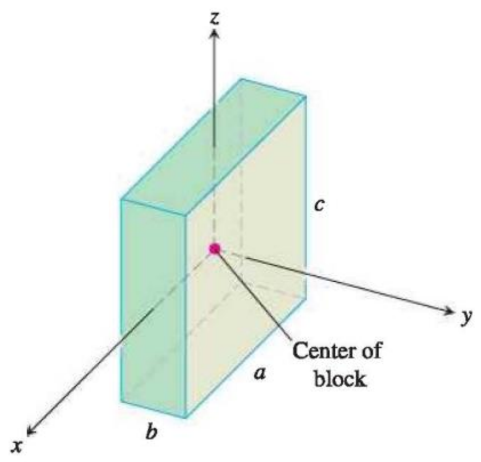
$$= \frac{7k}{12}.$$ First moment about yz-plane

So, $\bar{x} = \dfrac{M_{yz}}{m} = \dfrac{7k/12}{k} = \dfrac{7}{12}$

Finally, from the nature of ρ and the symmetry of x, y, and z in this solid region, you have $\bar{x} = \bar{y} = \bar{z}$, and the center of mass is $\left(\frac{7}{12}, \frac{7}{12}, \frac{7}{12}\right)$.

Masses and Moments in Three Dimensions

Example 3: Find I_x, I_y , I_z for the rectangular solid of constant density ρ shown in the below Figure



Solution The formula for I_x gives

$$I_x = \int_{-c/2}^{c/2} \int_{-b/2}^{b/2} \int_{-a/2}^{a/2} (y^2 + z^2)\, \delta \, dx\, dy\, dz.$$

We can avoid some of the work of integration by observing that $(y^2 + z^2)\delta$ is an even function of x, y, and z since δ is constant. The rectangular solid consists of eight symmetric pieces, one in each octant. We can evaluate the integral on one of these pieces and then multiply by 8 to get the total value.

$$I_x = 8\int_0^{c/2} \int_0^{b/2} \int_0^{a/2} (y^2 + z^2)\, \delta \, dx\, dy\, dz = 4a\delta \int_0^{c/2} \int_0^{b/2} (y^2 + z^2)\, dy\, dz$$

$$= 4a\delta \int_0^{c/2} \left[\frac{y^3}{3} + z^2 y\right]_{y=0}^{y=b/2} dz$$

$$= 4a\delta \int_0^{c/2} \left(\frac{b^3}{24} + \frac{z^2 b}{2}\right) dz$$

$$= 4a\delta \left(\frac{b^3 c}{48} + \frac{c^3 b}{48}\right) = \frac{abc\delta}{12}(b^2 + c^2) = \frac{M}{12}(b^2 + c^2). \qquad M = abc\delta$$

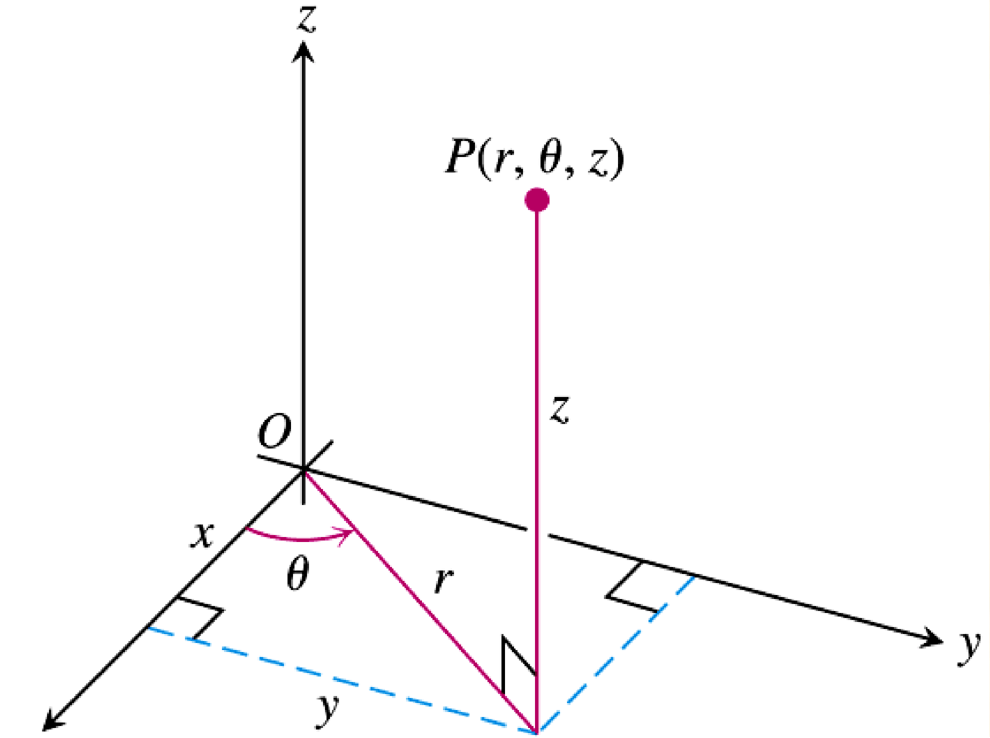
Triple Integrals in Cylindrical and Spherical Coordinates

DEFINITION Cylindrical Coordinates

Cylindrical coordinates represent a point P in space by ordered triples (r, θ, z) in which

1. r and θ are polar coordinates for the vertical projection of P on the xy-plane
2. z is the rectangular vertical coordinate.

Triple Integrals in Cylindrical and Spherical Coordinates



Equations Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$

$$r^2 = x^2 + y^2, \qquad \tan\theta = y/x$$

Triple Integrals in Cylindrical and Spherical Coordinates

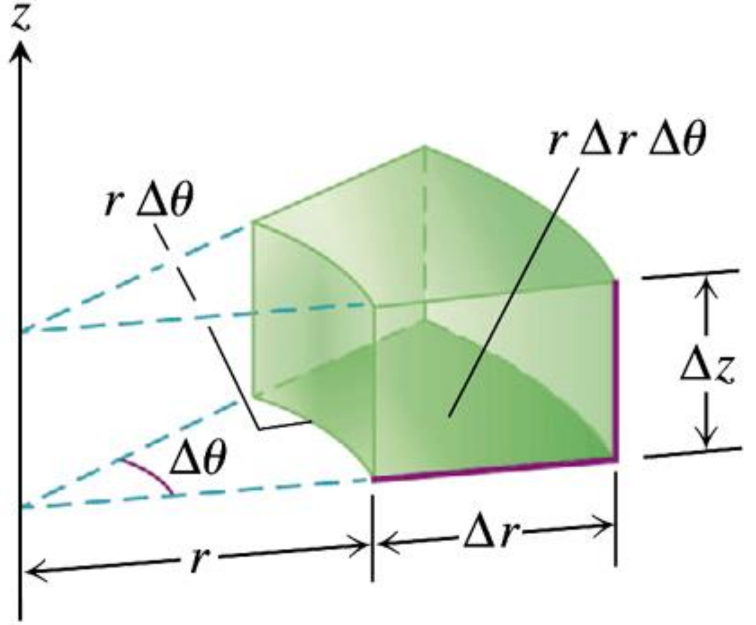


FIGURE 15.38 In cylindrical coordinates the volume of the wedge is approximated by the product $\Delta V = \Delta z\, r\, \Delta r\, \Delta\theta$.

Triple Integrals in Cylindrical and Spherical Coordinates

DEFINITION Spherical Coordinates

Spherical coordinates represent a point P in space by ordered triples (ρ, ϕ, θ) in which

1. ρ is the distance from P to the origin.
2. ϕ is the angle $\overrightarrow{OP}$ makes with the positive z-axis $(0 \le \phi \le \pi)$.
3. θ is the angle from cylindrical coordinates.

Triple Integrals in Cylindrical and Spherical Coordinates

Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates

$$\begin{aligned} r &= \rho \sin \phi, \qquad & x &= r \cos \theta = \rho \sin \phi \cos \theta, \\ z &= \rho \cos \phi, \qquad & y &= r \sin \theta = \rho \sin \phi \sin \theta, \end{aligned} \tag{1}$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}.$$

Triple Integrals in Cylindrical and Spherical Coordinates

Coordinate Conversion Formulas

Cylindrical to Rectangular	Spherical to Rectangular	Spherical to Cylindrical
$x = r\cos\theta$	$x = \rho\sin\phi\cos\theta$	$r = \rho\sin\phi$
$y = r\sin\theta$	$y = \rho\sin\phi\sin\theta$	$z = \rho\cos\phi$
$z = z$	$z = \rho\cos\phi$	$\theta = \theta$

Corresponding formulas for dV in triple integrals:

$$\begin{aligned} dV &= dx\,dy\,dz \\ &= dz\,r\,dr\,d\theta \\ &= \rho^2\sin\phi\,d\rho\,d\phi\,d\theta \end{aligned}$$

Thanks a lot …